GIHAN LAKMAL

AI Engineer

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Github - https://github.com/Gihan007 | Portfolio - https://gihan007.github.io/my-portfolio

PROFESSIONAL SUMMARY

AI/ML Engineer and researcher with nearly two years of experience designing, building and deploying agentic AI, multi agent, machine learning and generative AI solutions. Specializes in autonomous agent architectures, LangChain and LangGraph orchestration, tool and function calling, Retrieval Augmented Generation, LLM driven workflow automation, NLP, computer vision and time series forecasting. Delivered production oriented systems for financial intelligence, healthcare voice automation, insurance document extraction and workplace safety, with end to end ownership across model integration, FastAPI back ends, databases, Docker, cloud deployment, QA, monitoring and performance optimization. BSc (Honours) Computer Science graduate with an AI specialization and a 3.61 GPA.

PROFESSIONAL EXPERIENCE

AI Engineer **Jul 2026 - Present**

VInvest Pvt Ltd - Full Time

- Architected a fintech multi-agent AI platform on LangGraph with coordinator, planner, supervisor and specialist agent nodes, using stateful memory with Postgres checkpointing and autonomous task routing for deep financial analysis.
- Took it from idea to live product through DevOps practices, with containerized, cloud-deployed, scalable services I owned end to end.

Co Founder - AI Research & Engineering **Dec 2025 - Present**

ArcheAI - Part Time

- Co founded an AI research and engineering startup developing agentic systems for enterprise and client automation, including an agentic WhatsApp bot, insurance fax extraction models and a voice call agent platform.
- Designed complete agent architectures combining LLM orchestration, autonomous task execution, tool integration and workflow automation.
- Applied RAG, semantic embedding optimization and hybrid neuro symbolic workflows to improve contextual reasoning, reliability and production readiness.
- Contributed across solution architecture, backend engineering, model integration, testing, deployment and client delivery.

AI/ML Trainee Associate Engineer **Jul 2025 - Jul 2026**

SenzMate AIoT

- Designed modular AI agent nodes and autonomous workflow pipelines for the Interplay Autonomous Platform, enabling multi agent task routing, memory management and decision making.
- Built finance focused multi agent solutions using LangChain, LangGraph and modern LLM orchestration frameworks with custom tools and RAG pipelines.
- Delivered client projects with end to end ownership covering agent architecture, AI model integration, backend development, deployment and production optimization.
- Supported scalable API integration, testing and deployment using FastAPI, Docker and cloud infrastructure.

AI Research Assistant **Nov 2024 - Jun 2025**
AI Research Environment

- Worked on generative AI models for Sinhala and Tamil, collecting and processing a 40 million word bilingual corpus to improve relevance for Sri Lankan language users.
- Applied generative AI, computer vision and audio generation techniques to native language research problems.
- Used parameter efficient fine tuning approaches including PEFT, LoRA and QLoRA for language model adaptation and experimentation.

EDUCATION

- **BSc (Honours) Computer Science - Artificial Intelligence Specialization** **Jul 2022 - Jul 2026**
  University of Kelaniya, Sri Lanka - GPA 3.61
- **Bachelor of Information Technology - External Programme** **Jul 2021 - Aug 2022**
  University of Colombo, Sri Lanka
- **G.C.E. Advanced Level - Physical Science Stream** **2020**
  Nalanda College, Colombo 10 - Z Score 1.563

CORE COMPETENCIES & SKILLS

- Agentic AI Architecture
- Multi Agent Systems
- Autonomous Task Execution
- Supervisor Graphs
- Tool & Function Calling
- Model Context Protocol
- Retrieval Augmented Generation
- LLM Fine Tuning
- Prompt Engineering
- Generative AI
- Natural Language Processing
- Computer Vision
- Time Series Forecasting
- FastAPI Development
- Vector Databases
- Knowledge Graphs
- MLOps
- Model Monitoring
- Client Solution Delivery

TECHNICAL SKILLS

- **Agentic AI & Orchestration** - AI agents, multi agent systems, LangChain, LangGraph, ReAct, supervisor graphs, tool/function calling, MCP, LangSmith and Temporal
- **AI & Data Science** - Machine learning, deep learning, NLP, generative AI, RAG, transformers, prompt engineering, computer vision and time series forecasting
- **Programming & Application Development** - Python, Java, JavaScript, C, C#, React, Next.js, HTML and CSS
- **Frameworks, Databases & Platforms** - FastAPI, TensorFlow, Keras, Scikit learn, Pandas, NumPy, ChromaDB, Neo4j, PostgreSQL, MongoDB and Redis
- **MLOps & DevOps** - Docker, Kubernetes, GKE, CI/CD, MLflow, DVC, Prometheus, Grafana, Jenkins, model monitoring and logging
- **Tools** - Git, GitHub, Jupyter Notebook, VS Code, Anaconda and Google Cloud Platform

SELECTED AI & MACHINE LEARNING PROJECTS

VInvest AI Valuation Intelligence Engine
Multi Agent Financial AI Platform - Client Project

- Built the AI decision engine for a financial valuation platform in which autonomous agents interpret user queries, classify company types, retrieve financial signals, select valuation frameworks and execute valuation logic end to end.

- Implemented a supervisor graph architecture with ReAct style reasoning and Deep Research / Deep QA workflows coordinating specialist agents for retrieval, reasoning, valuation and automated report generation.
- Integrated custom tool and function calling across PostgreSQL, MongoDB, Neo4j and Redis.
- Owned QA, load testing and deployment validation using Docker, Swagger, live API testing and GKE across multiple ticker categories.
- Technologies - Python, FastAPI, LangChain, LangGraph, Redis, ChromaDB, Neo4j, PostgreSQL, MongoDB, RAG, MCP, Docker and GKE

Advanced Agentic Hybrid OCR - Prior Authorization Fax Intelligence
Production Document Processing System - ArcheAI Client Project

- Designed and deployed an agentic OCR system for autonomous extraction from complex multi page insurance approval faxes, achieving 94-98% extraction accuracy.
- Combined regex extraction, LLM validation and correction, vision language model gap filling and confidence based human in the loop review.
- Built a seven phase asynchronous pipeline with job tracking, extraction storage, audit logging and multi tenant support.
- Optimized the production workflow to process more than 100 documents per hour.
- Technologies - Python, FastAPI, Temporal, PostgreSQL, OpenAI Vision Models, Tesseract, PaddleOCR, React and Docker

AI Medical Appointment Voice Agent Platform
Autonomous Healthcare Voice Agent - USA Client

- Built an autonomous voice agent for live appointment calls, routing conversations through greeting, availability, rescheduling, cancellation, general Q&A and wrap up states.
- Integrated real time speech to text, LLM reasoning, clinic specific RAG and text to speech with voice cloning.
- Improved responsiveness through fast path intent handling and response caching.
- Technologies - LangGraph, FastAPI, WebSocket Streaming, Deepgram STT, Gemini, Groq, RAG, Voice Cloning and Prompt Engineering

Construction Worker Safety Monitoring System
Computer Vision Safety Compliance MVP - Indian Client

- Built a computer vision MVP detecting workers, helmets and safety vests from images and video.
- Applied safety compliance rules to classify safe and unsafe conditions and generate annotated outputs.
- Technologies - YOLOv8, Ultralytics, Python, FastAPI, OpenCV, Docker and PostgreSQL

Medical Drug Sales Prediction by Area in Sri Lanka
IEEE Research Publication - Time Series Forecasting

- Built a production oriented weekly, area level pharmaceutical sales forecasting and inference pipeline using an ensemble of deep learning and statistical models.
- Added SHAP explain ability, causal and counterfactual analysis, containerized inference, CI tests and drift monitoring.
- Technologies - Python, GRU, LSTM, Transformer, TFT, N BEATS, LightGBM, XGBoost, Prophet, SHAP, Docker, Flask and FastAPI

LANGUAGES

- English - Professional Proficiency
- Sinhala – Native